

LUT School of Energy Systems

Degree Programme

**HDD4060 Technical Computing Software (HDD) - Contact teaching**

# Research Report on Characteristics in RLC Series Circuit through Numerical Simulation

## Abstract

This study takes the RLC series circuit as the research object and uses theoretical derivation combined with Simulink numerical simulation to analyze three key characteristics: current amplification, damping, and phase shift. Based on Kirchhoff's Voltage Law, a second-order differential equation is established to distinguish transient and steady-state responses, and the laws of resonance, damping, and phase variation are clarified. Multi-parameter simulations are carried out using the Runge-Kutta (ode45) algorithm. The results show that the simulated waveforms and values are in good agreement with theoretical calculations: the current is significantly amplified at resonance, and a larger resistance leads to faster damping attenuation. The simulation involves numerical discretization and reading errors; the model does not consider component parasitic effects and nonlinearity, and the beat phenomenon exists. Through the application of numerical simulation technology in the RLC series circuit, this study helps us better master and understand numerical simulation methods.

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## 1. Introduction

A series RLC circuit is an electrical system composed of a resistor (R), an inductor (L), and a capacitor (C) connected in series, ensuring that the same current flows through each component. The dynamic response of the circuit depends on the interplay between resistance, inductance, and capacitance, leading to important phenomena such as resonance, damping, and phase shift when driven by an alternating voltage source. (Hayt et al., 2019)

RLC circuits are widely and importantly applied in the fields of frequency selection and tuning, signal filtering, sinusoidal oscillation, impedance matching, etc. The applications of RLC circuits are widely discussed in today's academic community. According to Alexander and Sadiku (2016), the transient and steady-state responses of such circuits are critical for designing stable power filters and resonance-based communication devices. While analytical methods offer exact solutions for ideal models, Gómez (2013) emphasizes that numerical simulation provides a more robust framework for validating these equations against complex, real-world transients. Furthermore, the application of the Dormand-Prince algorithm (the basis for MATLAB's ode45) has become a standard in computational physics for solving such dynamic systems with high precision and adaptive step-sizing (Dormand & Prince, 1980).

In this project, we first theoretically derived the current amplification characteristics, damping characteristics, and the phase shift characteristics between current and voltage of RLC circuits through formulas. Then we used MATLAB and Simulink for computer simulation. These simulations were based on the numerical method known as the Runge-Kutta method. After the simulation, we compared the obtained results with the theoretical results and found that they were basically consistent, which met the expectations.

By exploring the working principles of RLC circuits, this project will help us understand the principles behind phenomena in daily life such as wireless charging and induction heating, and to better understand the significance of applying numerical simulation methods in practical engineering projects.

## 2. Methods

### 2.1 Mathematical Model of an RLC Series Circuit

In an RLC series circuit, according to Kirchhoff's Voltage Law (KVL), the circuit satisfies the following differential equation:

$$L\frac{di}{dt} + Ri + \frac{1}{C}\int idt = V(t) \tag{1}$$

This equation is used to describe the dynamic response of the circuit, where L (H) stands for inductance, R (Ω) stands for resistance, and C (F) stands for capacitance. t (s) represents time. i is the current in the circuit as a function of time t, measured in amperes (A). V(t) is the total

voltage of the circuit as a function of time t, measured in volts (V) (MathWorks, 2024a).

In this project, the random initial values of our group are as follows: Resistance (Ω) = 64, Capacitance (mF) = 471.6, Inductance (mH) = 96, Voltage (V): 120 (RMS), Frequency (Hz): 60. And the total voltage applied to the circuit is a sinusoidal alternating voltage, and its function expression is as follows:

$$V(t) = U_m sin(\omega t) \tag{2}$$

where $\omega$ (rad/s) stands for angular frequency of sinusoidal alternating voltage.

## 2.2 Theoretical Explanation of The Three Characteristics of Electric Current

By differentiating both sides of Equation (1) with respect to time t, the following equation can be obtained:

$$L\frac{d^2i}{dt^2} + R\frac{di}{dt} + \frac{1}{C}i = U_m \omega cos(\omega t) \tag{3}$$

This is a second-order linear nonhomogeneous differential equation (MathWorks, 2024a; Hayt et al., 2019). The structure of its solution consists of two parts: one is called the transient solution, and the other is called the steady-state solution. The final solution is the algebraic sum of these two parts, which can be expressed by the following equation:

$$i(t) = i_h(t) + i_p(t) \tag{4}$$

$i_h(t)$ is the transient solution and $i_p(t)$ is the steady-state solution. (Boylestad, 2013)

The transient solution can be obtained by the characteristic equation method. According to the form of Equation (3), the characteristic equation is obtained as follows:

$$Lr^2 + Rr + \frac{1}{C} = 0 \tag{5}$$

First, define the damping coefficient $\delta = \frac{R}{2L}$ and natural angular frequency $\omega_0 = \frac{1}{\sqrt{LC}}$. Then the solution of Equation (5) is

$$r = \frac{-R \pm \sqrt{R^2 - \frac{4L}{C}}}{2L} = -\delta \pm \sqrt{\delta^2 - \omega_0^2} \tag{6}$$

Now it needs to be divided into three situations: (Hayt et al., 2019)

When $\delta > \omega_0$,

$$i_h(t) = Ae^{r_1 t} + Be^{r_2 t} \tag{7}$$

When $\delta = \omega_0$,

$$i_h(t) = (A + Bt)e^{-\delta t} \tag{8}$$

When $\delta < \omega_0$,

$$i_h(t) = e^{-\delta t}\left(Acos\sqrt{\omega_0^2 - \delta^2}t + Bsin\sqrt{\omega_0^2 - \delta^2}t\right) \tag{9}$$

The steady-state solution can be solved using the complex number method (vector method) (Boylestad, 2013), and the final steady-state solution can be obtained as:

$$i_p(t) = \frac{U_m}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} sin(\omega t - \varphi) \tag{10}$$

$$\varphi = arctan\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right) \tag{11}$$

The characteristics of current amplification in the circuit can be explained by Equation (10). Through Equation (10), it can be found that the amplitude of the current is affected by the relative magnitudes of the inductance and capacitance values. When

$$\omega L = \frac{1}{\omega C} \tag{12}$$

At this time, the angular frequency of the sinusoidal AC voltage source is equal to the resonant angular frequency of the circuit, that is, the frequency of the voltage source $f$ (Hz) is equal to the resonant frequency of the circuit:

$$f = \frac{1}{2\pi\sqrt{LC}} \tag{13}$$

That means when the circuit resonates, the amplitude of the current reaches its maximum value. Therefore, the current amplification phenomenon will occur in the circuit near the resonant frequency. (Hayt et al., 2019)

Next, the damping characteristics in the circuit will be explained. Through observing the three forms of the transient solutions of the current [Equation (7), Equation (8), Equation (9)], it is not difficult to find that:

$$i_h(t) \sim e^{-f(R)t} \tag{14}$$

f(R) is always a monotonically increasing function. Therefore, the larger R is, the faster the transient solution of the current attenuates, and the whole current will tend to the steady-state solution more quickly. It can also be found from Equation (10) that when R is very large, the amplitude of the steady-state solution of the current will become very small. The essence of these phenomena is that resistance causes energy dissipation in the system, which gradually weakens the oscillation. (Hayt et al., 2019)

The phase difference between voltage and current in the circuit can be reflected by Equation (11). Its core influencing factor is still the relative magnitude of inductance and capacitance. When

$$\omega L > \frac{1}{\omega C} \tag{15}$$

$\varphi > 0$, the circuit is inductive, and the phase of the current lags behind that of the voltage. When

$$\omega L < \frac{1}{\omega C} \tag{16}$$

$\varphi < 0$, the circuit is capacitive, and the phase of the current leads that of the voltage. When

$$\omega L = \frac{1}{\omega C} \tag{17}$$

$\varphi = 0$, The current and voltage are in phase, and resonance occurs in the circuit. (Boylestad, 2013)

## 2.3 Numerical Methods and Simulation

In this project, modeling is carried out using Simulink, an additional tool of MATLAB, combined with the Simscape Electrical module library (MathWorks, 2024c). The models used include Sine Wave, Controlled Voltage Source, Resistor, Capacitor, Inductor, Electrical Reference, Current Sensor, Voltage Sensor, Scope and Solver Configuration. The schematic diagram of the circuit modeling is as follows.

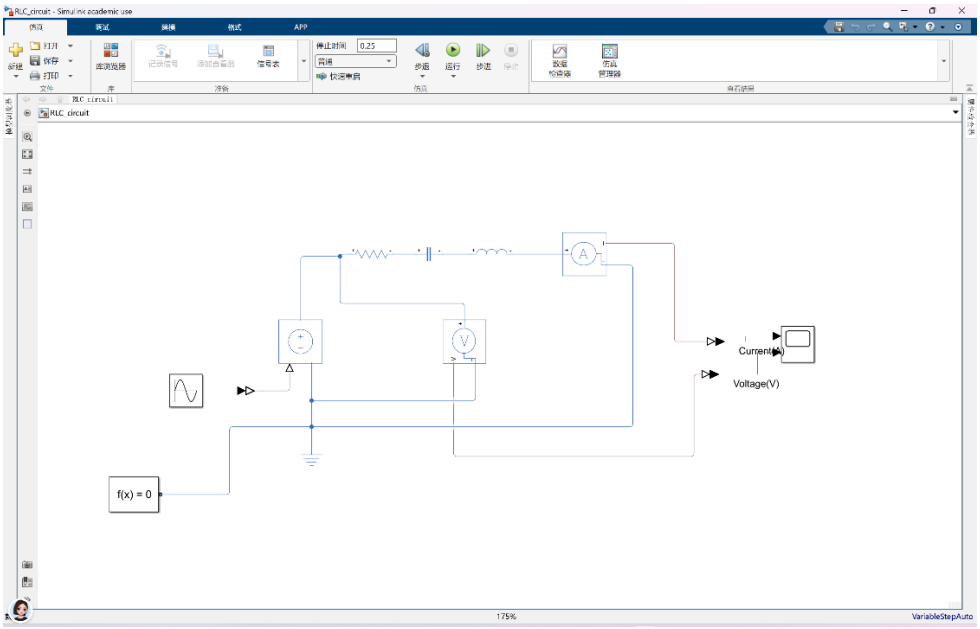



Fig.1 The schematic diagram of the circuit modeling

Simscape automatically converts the circuit into a system of differential equations and solves it using the numerical solver ode45 (based on the Runge-Kutta 4(5) method). (MathWorks, 2024b; Chapra & Canale, 2015)

The Runge-Kutta method is a numerical method for solving ordinary differential equations. It improves the accuracy of the solution by calculating the slopes at multiple points within each time step and taking a weighted average. Compared with the Euler method, which uses only a single slope, this method can accurately describe the changing trend of the solution. (Chapra & Canale, 2015) The ode45 function in MATLAB adopts adaptive step-size control, which automatically adjusts the step size according to the variation of the solution. (MathWorks, 2024b). This improves computational efficiency while ensuring accuracy, making it suitable for solving the RLC circuit problems in this project.

During the simulation, we assumed that the initial current is zero, change some parameters of the circuit (such as R, L, C), and obtained some regular curves through the simulation model to verify the theoretical characteristics of the current. Since the initially given parameters were not easy to visualize in the simulation, we modified the initial parameters in the simulation. Although these simulations used off-the-shelf modules in Simulink, they were still based on numerical integration methods, which were consistent with theoretical solutions.

## 3. Results

All simulations in this project were carried out with the RMS voltage (120 V) and frequency (60 Hz) of the sinusoidal AC voltage source kept constant, while only the values of resistance R(Ω), capacitance C(F), and inductance L(H) were changed. Six sets of initial parameters were set in the simulation, and each case corresponds to one graph. The simulation time T(s) may vary for different cases.

**Case 1.** R=5, C=0.01, L=0.015, T=0.25. The graph is shown in Figure 2.

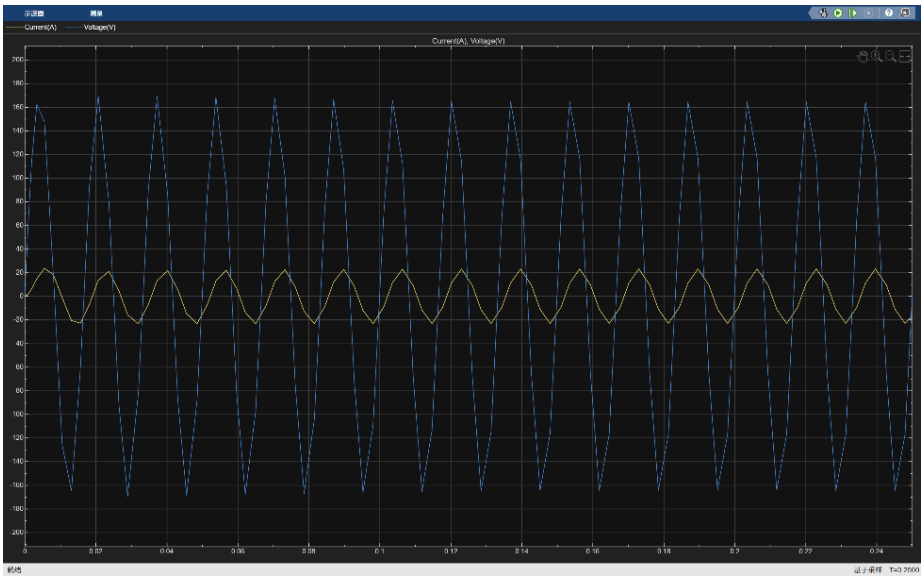



Fig.2 The graph for **Case 1**

**Case 2.** R=5, C=0.0002, L=0.015, T=0.25. The graph is shown in Figure 3.

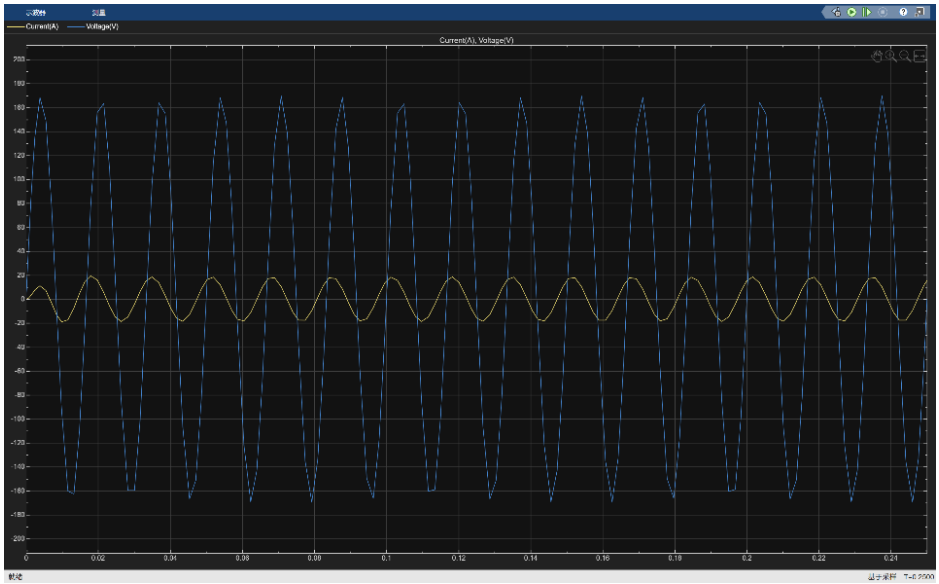



Fig.3 The graph for **Case 2**

**Case 3.** R=5, C=0.01, L=0.0007, T=0.25. The graph is shown in Figure 4.

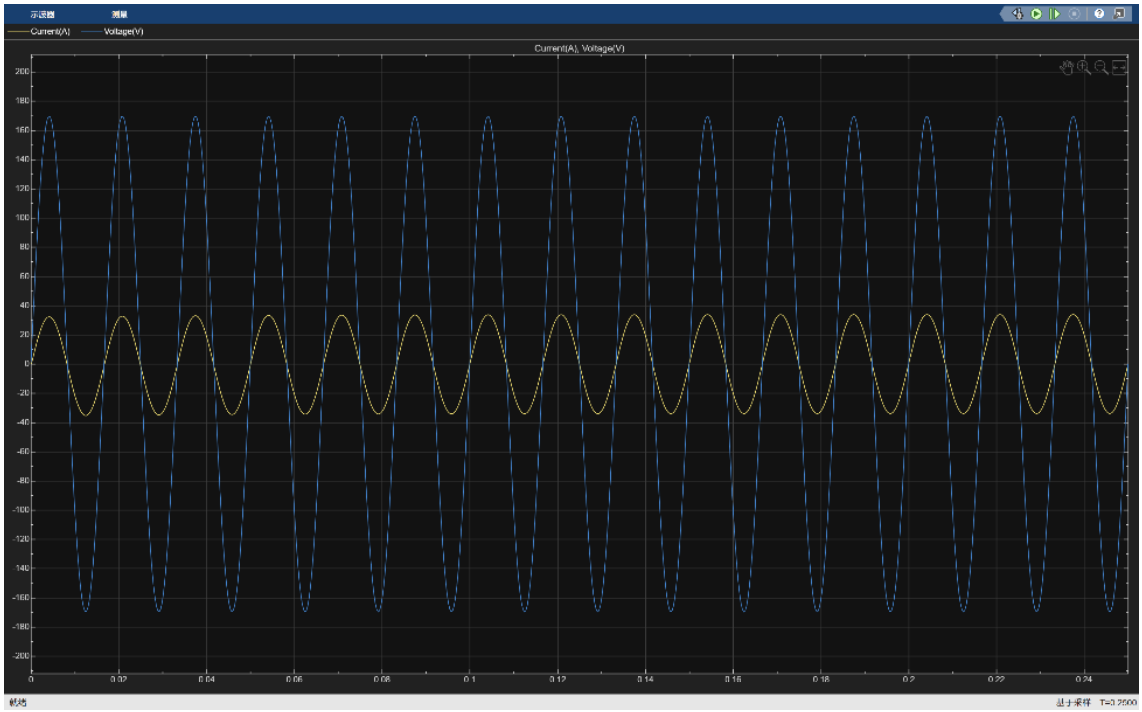



Fig.4 The graph for **Case 3**

**Case 4.** R=0.05, C=0.01, L=0.015, T=1. The graph is shown in Figure 5.

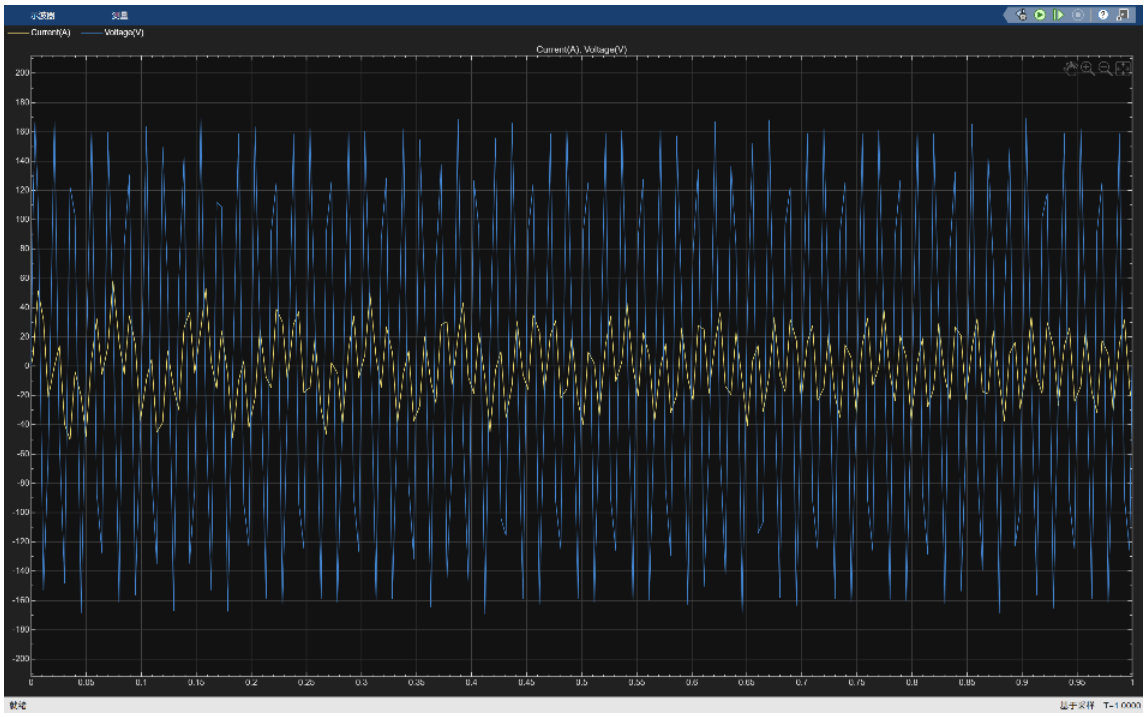



Fig.5 The graph for **Case 4**

**Case 5.** R=0.1, C=0.01, L=0.015, T=1. The graph is shown in Figure 6.

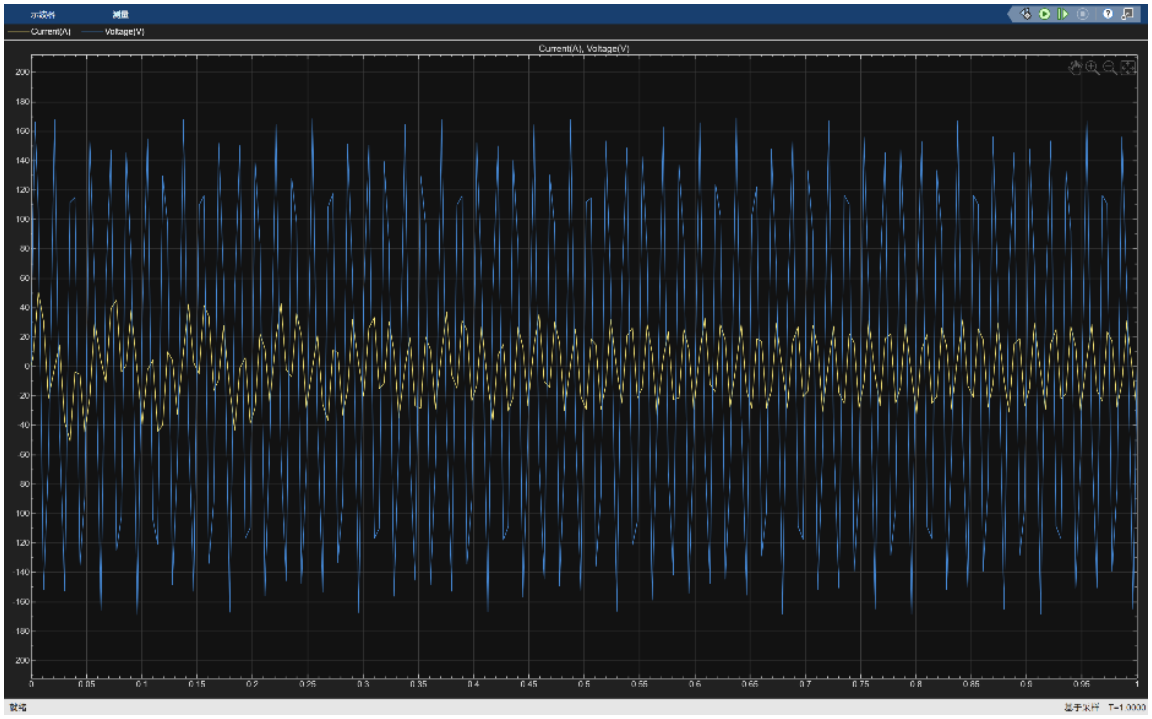



Fig.6 The graph for **Case 5**

**Case 6.** R=0.5, C=0.01, L=0.015, T=1. The graph is shown in Figure 7.

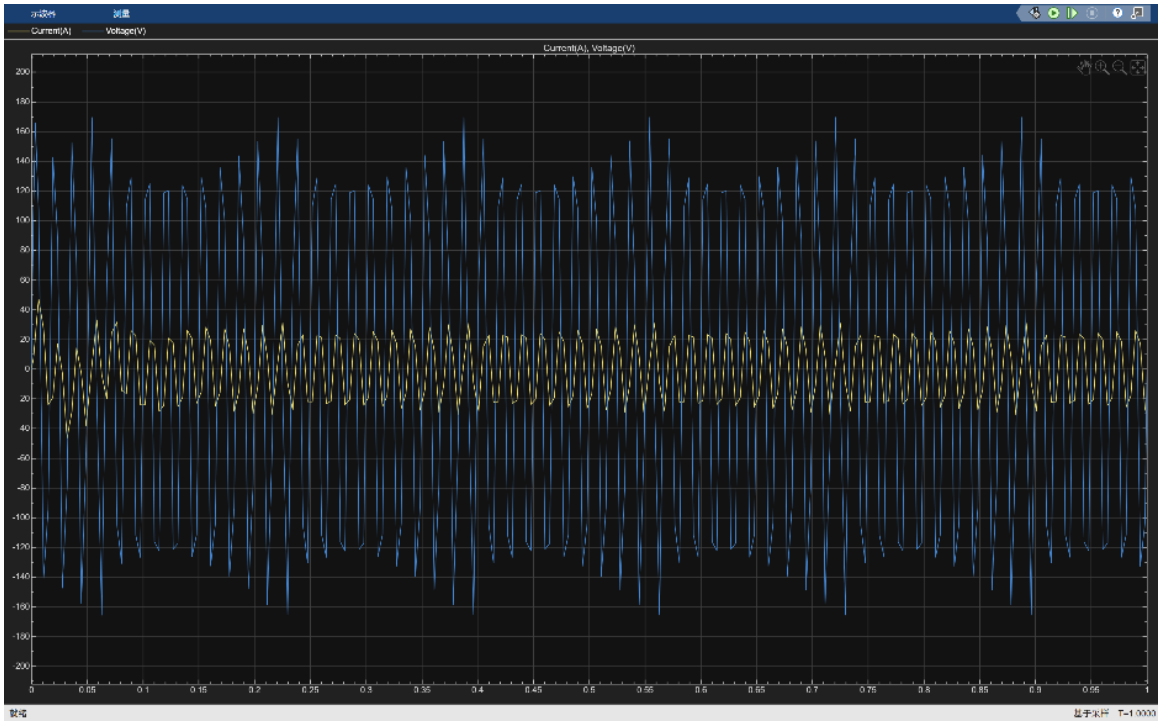



Fig.7 The graph for **Case 6**

## 4. Analysis

### 4.1 Simulation of Phase Shift and Amplification Phenomena

Case 1, Case 2 and Case 3 correspond to the simulations of these phenomena. Since both phenomena can be reflected by changing the values of $C$ and $L$, $R$ remains constant in these three cases.

In Case 1, it can be found that $\varphi > 0$ [Equation (11)] through calculation. Theoretically, the circuit is inductive at this time, and the current phase should lag behind the voltage phase. The simulation results shown in Figure 2 are consistent with theoretical expectations. In Case 2, it can be found that $\varphi < 0$ [Equation (11)] through calculation. Theoretically, the circuit is capacitive at this time, and the current phase should lead the voltage phase. The simulation results shown in Figure 3 are consistent with theoretical expectations. In Case 3, it can be found that $\varphi \approx 0$ [Equation (11)] through calculation. Theoretically, resonance occurs in the circuit at this time, and the current phase should be in phase with the voltage phase. In summary, all of the above three simulation results reflect well the real phase shift characteristics of the series RLC circuit under different initial conditions.

In Case 1, the simulated peak current under steady state is approximately 23.005 A, while the corresponding theoretical peak current is about 23.084 A [Equation (10)]. In Case 2, the simulated peak current under steady state is approximately 18.628 A, while the corresponding theoretical peak current is about 18.641 A [Equation (10)]. In Case 3, the simulated peak current under steady state is approximately 33.610 A, while the corresponding theoretical peak current is about 33.941 A [Equation (10)]. From these data, it is not difficult to find that the simulated peak current is close to the theoretical peak current, indicating that the results obtained by the numerical simulation method are consistent with theoretical expectations. It can also be seen that both the theoretical and simulated values are larger than the previous two values when the circuit resonates. This indicates that this numerical simulation method well reproduces the process of generating current amplification by adjusting the values of $C$ and $L$ in the circuit to make its resonant frequency of the circuit close to the frequency of the sinusoidal AC voltage source [Equation (12), Equation (13)].

### 4.2 Simulation of Damping Phenomena

Case 4, Case 5 and Case 6 are simulations of this phenomenon. Since damping is mainly caused by resistance, the values of $C$ and $L$ will remain unchanged in these three cases, and only the value of $R$ will be varied.

Comparing the waveforms corresponding to the three cases, it is not difficult to observe that the peak value of the current oscillation is not stable and uniform, but decays continuously from a relatively large value and finally approaches a steady value. Moreover, the smaller the value of R, the longer it takes for the current to stabilize. This is consistent with the theoretical expectation of Equation (14), indicating that the numerical simulation clearly reflects the

damping characteristics caused by resistance in the circuit.

## 5. Discussion and Conclusions

### 5.1 Error Analysis

Although the numerical simulation in this project well reflects the theoretical characteristics of the series RLC circuit supplied by a sinusoidal AC voltage source, errors between the simulated results and the theoretical results are inevitable due to the inherent approximation of numerical methods. First, there is the numerical error generated by Simulink. Numerical errors are caused by the discretization of continuous differential equations. The Runge-Kutta solver computes approximate solutions using discrete time steps, and the accuracy of the results depends on the solver's step size and tolerance settings, which leads to slight deviations from the theoretical solution. In addition, since the analysis of the simulation results is all derived from the graphs, the resolution of the graphs themselves and the accuracy of reading will both cause errors in the results.

### 5.2 Model Limitations and Further Research

The series RLC circuit investigated in this project is an ideal circuit model. This model assumes that all components are ideal and free from parasitic effects, such as the internal resistance in non-ideal inductors or leakage in non-ideal capacitors. Meanwhile, the model assumes that all components are linear, ignoring the nonlinear characteristics of components under certain conditions. The circuit is assumed to be in an ideal environment, unaffected by factors such as temperature and electromagnetic interference. In addition, there is a structural issue in the circuit modeling in Simulink for this project. Under this connection configuration, the voltage sensor measures the total voltage across the series combination of the resistor, capacitor, and inductor, rather than the input voltage of the controlled voltage source. This results in the peak value of the simulated AC voltage not being stable but oscillatory, which is known as the Beat Phenomenon. This phenomenon becomes more pronounced when R is small (underdamped), whereas it disappears entirely when R is large (overdamped) (Hayt et al., 2019). This also accounts for the unstable voltage peaks observed in the six result graphs.

Future research directions can start from the causes of errors or the limitations of the model. By optimizing the model, such as adjusting the time step of the solver to obtain more accurate simulation results, considering the actual conditions of components in certain environments to improve the practical feasibility of the model, and incorporating the beat phenomenon into the model system to more comprehensively simulate the characteristics of the series RLC circuit, and so on. In addition, this project mainly qualitatively reflects some characteristics of the series RLC circuit through numerical simulation. Future research can also focus on quantitative studies of a specific phenomenon, such as simulating the exact value of phase shift to better understand the characteristics of numerical simulation in reflecting actual phenomena.

## 5.3 Engineering Significance

The research of this project not only demonstrates the characteristics of the series RLC circuit supplied by a sinusoidal AC voltage source, but also reflects the application value of numerical simulation methods for the visualization of physical phenomena. Numerical simulation can effectively reflect the operation process of practical engineering projects at a low cost, which is of great significance for promoting the efficient progress, optimization and innovation of engineering projects.

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